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Girls Who Code + RTX Data Science & AI Study

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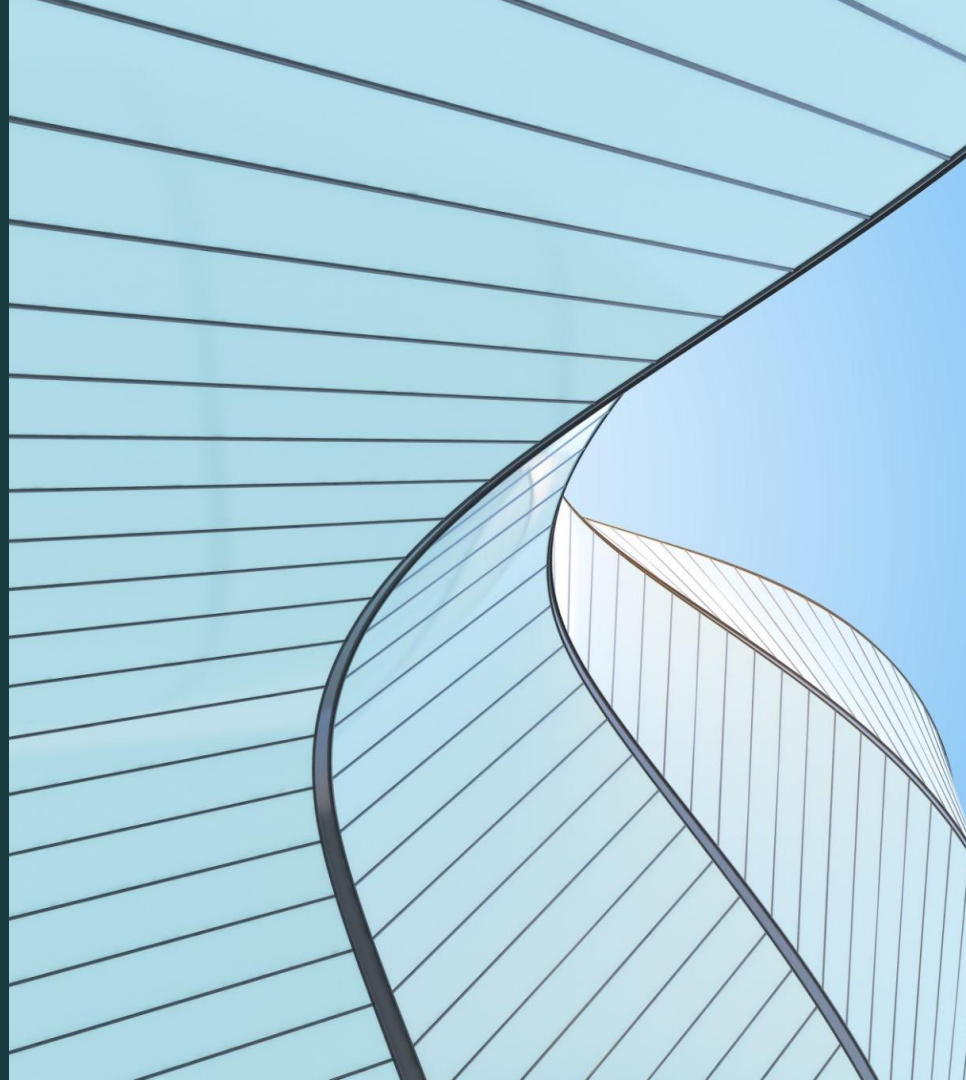


Problem Statement

We look to solve weather related issues, aiming to fix maintenance scheduling and catching the issues sooner, ensuring the safety of both the passengers and the schedule.

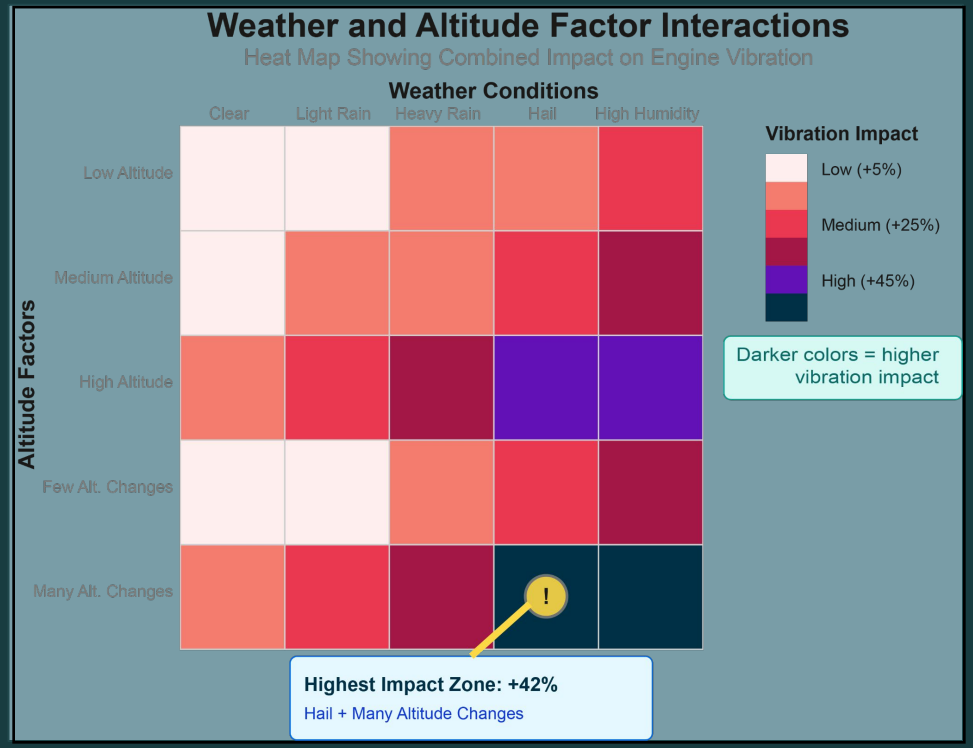
Key Findings

If vibration > 2.5 mm/s →

If precipitation > 70mm AND humidity > 80% →

THEN, scheduling needs to happen within 40 days →

Graph to demonstrate the effects the weather can have on the vibration on the engine with other factors



What do we do with these findings?

Recommendations

Get the **enhanced vibration sensors** to catch the vibration sooner, thus ensuring the safety of passengers and schedule (if there is an emergency, everyone loses).

Get the **Weather Data Integration** to plan ahead of time and save the cost of an emergency and rescheduling.

Both are costly, but within budget and will save the company money in the long term.

ROI Calculation

Cost Calculation
Weather Data Integration = $ 180,000

Enhanced Vibration Sensors = $ 320,000

Total Cost = $ 500,000

ROI = Annual Benefit - Implementation Cost

Total Annual Benefit: $2.4M

ROI = $2.4M - $ 500,000 = **$ 1.9 M**

Investor Segment

Implementation

Phase 1: Buying and shipping components

Estimated time: 1-2 months

This is where the company is losing the most money, but to invest in a better future

Phase 2: Installing components

Estimated time: 1-2 months

This is where the plan is at most risk for failure, if the parts are not correctly assembled and implemented. Time will be taken for testing to ensure success.

Phase 3: Testing

Estimated time: 3-4 months

Testing the new systems for accuracy and reliable is vital, and a point in the process where if it fails, we can perhaps refund or consider other options

Phase 4: Break Even

Estimated time: around 4 months in

The costs saved from emergency repairs and rescheduling evens out the original cost

Phase 5: Benefit

Months+

After even is broken, from months on the costs saved will be of benefit and the investment will have been worth it

Thank you

Conclude the presentation by thanking your audience for listening and participating.